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Continuous clean centrifuge

Abstract

A continuous clean centrifuge is provided for separating denser materials from a flow of liquid. The device includes a manifold assembly rotatably supported within a frame assembly and driven by a motor via a belt and sprocket system. A first rotary coupler connects a manifold inlet to the source of liquid, and a second rotary coupler connects a manifold outlet to downstream usage. A plurality of valve inlets and remotely-operated release valves selectively discharge separated denser materials during operation. A main logic controller regulates motor speed between a first rotation state for separation and a second rotation state for purging, based on inputs from a programming interface, variable frequency drives, and multiple sensors. A shroud encloses the frame assembly to provide environmental protection. The centrifuge provides continuous, automated separation of denser materials from a liquid flow.

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Background/Summary

RELATED APPLICATIONS
Field of the Device
(1) The present device relates generally to fluid filtration and separation devices, and more particularly to a continuous centrifuge system configured to separate denser materials from a flowing liquid stream through automated, programmable rotation and discharge cycles.

Background of the Device

(2) In many industries, including water treatment, food and beverage production, oil refining, agricultural processing, and industrial fluid management, it is often necessary to remove suspended solids, particulates, and other dense contaminants from a flowing liquid. Traditional separation methods such as settling tanks, membrane filtration, and conventional centrifuges each present distinct drawbacks. Settling systems require large footprints and long processing times, while membranes are prone to clogging and require frequent cleaning or replacement. Conventional centrifuges, while effective at separating denser materials, often operate on a batch basis and require manual shutdowns to remove accumulated solids. Such interruptions reduce overall system throughput, increase labor and maintenance costs, and may expose the system to contamination risks upon reassembly.

(3) Additionally, existing centrifuge systems generally lack flexibility to adapt to varying types of liquids and contamination profiles without significant manual intervention. Many systems offer limited control over operational parameters such as rotational speed, discharge timing, and flow pressure, making it difficult to optimize performance across different service conditions. Moreover, the need to halt operations for periodic cleaning or purging limits the feasibility of continuous flow applications where uptime is critical, such as municipal water systems, food-grade liquid production, and high-volume industrial processes.

(4) Therefore, there exists a need in the art for an improved centrifuge system that can continuously process a flow of liquid without interruption, that can automatically separate and discharge denser materials, and that provides programmable, dynamic control of operational variables to optimize performance for a range of different fluids and use conditions. The continuous clean centrifuge of the present application fulfills these needs by providing an integrated, programmable system that separates and purges denser materials in a continuous flow environment in a highly efficient, reliable, and cost-effective manner, thereby overcoming the limitations of prior separation technologies without introducing complexity or excessive maintenance burdens.

SUMMARY OF THE DEVICE

(5) The present device provides a continuous clean centrifuge that addresses the limitations of prior systems by enabling uninterrupted flow-through operation while automatically separating and removing denser materials. The centrifuge includes a manifold assembly rotatably supported within a frame and driven by a motor coupled via a belt and sprocket, allowing the manifold assembly to be rotated relative to the frame assembly. A main logic controller regulates the motor to operate at a first rotation state to induce centrifugal separation of denser materials from a flow of liquid and at a second rotation state to discharge the collected materials through a plurality of remotely-operated release valves positioned along the manifold. The manifold assembly is fluidly connected to a source of the liquid via a first rotary coupler and directs a cleaned liquid stream through a second rotary coupler for downstream usage. A shroud encloses the frame assembly to protect the system from environmental conditions and to reduce noise and vibration during operation.

(6) The main logic controller interfaces with a programming interface, variable frequency drives, and a plurality of sensors to allow for either automated or manual control of the system. Operational parameters, including motor speed, discharge timing, and pump pressure, may be dynamically adjusted based on contamination levels, fluid properties, or system requirements. The continuous clean centrifuge is capable of processing a wide range of fluids, including but not limited to water, oils, sap, wines, beers, or compressed air treated as a flowable medium, and may be scaled to handle flows up to fifty million gallons per day. By allowing for continuous separation and selective purging without interruption of the flow, the system significantly improves process efficiency, reduces maintenance downtime, and provides a robust, scalable solution for industrial, agricultural, and commercial fluid processing applications.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The advantages and features of the present device will become better understood with reference to the following more detailed description and claims taken in conjunction with the accompanying drawings, in which like elements are identified with like symbols, and in which:

(2) FIG. **1** is a top front second side isometric view of the continuous clean centrifuge **10**, in accordance with a preferred embodiment of the present device;

(3) FIG. **2** is a bottom front first side isometric view of the continuous clean centrifuge **10**, in accordance with the preferred embodiment of the present device;

(4) FIG. **3** is a top plan view of the continuous clean centrifuge **10**, in accordance with the preferred embodiment of the present device;

(5) FIG. **4** is a top plan view of a lower manifold portion **32***b* of a manifold assembly **31** of the continuous clean centrifuge **10**, in accordance with the preferred embodiment of the present device;

(6) FIG. **5** is a side elevation view of the lower manifold portion **32***b* of the continuous clean centrifuge **10**, in accordance with the preferred embodiment of the present device;

(7) FIG. **6***a* is a pictorial view of the continuous clean centrifuge **10** operating in a first rotation state **11**, wherein denser material **75** from a flow of liquid **70** collects against an outer wall of the manifold **40** and valve inlet **41**, in accordance with an exemplary method of use;

(8) FIG. **6***b* is a pictorial view of the continuous clean centrifuge **10** operating in a second rotation state **12**, wherein denser material **75** is transferred away from the manifold **40** via a remotely-operated release valve **185**, in accordance with an exemplary method of use; and,

(9) FIG. **7** is an electrical schematic of a logic control for operating the continuous clean centrifuge **10**, in accordance with the preferred method of use of the present device.

DESCRIPTIVE KEY

(10) **10** continuous clean centrifuge **11** first rotation state **12** second rotation state **13** base plate **15** shroud **16** shroud front panel **17** shroud rear panel **18** shroud first side panel **19** shroud second side panel **20** frame assembly **21***a* first side upper horizontal segment **21***b* second side upper horizontal segment **22***a* front upper horizontal segment **22***b* rear upper horizontal segment **23***a* first side lower horizontal segment **23***b* second side lower horizontal segment **24***a* front lower horizontal segment **24***b* rear lower horizontal segment **25***a* first side front vertical segment **25***b* first side rear vertical segment **26***a* second side front vertical segment **26***b* second side rear vertical segment **27***a* first side front intermediate vertical segment **27***b* first side rear intermediate vertical segment **28***a* second side front intermediate vertical segment **28***b* second side rear intermediate vertical segment **30** manifold assembly **32***a* upper manifold portion **32***b* lower manifold portion **35** first rotary coupler **36** second rotary coupler **37** first bearing **38** second bearing **39** sprocket **40** manifold **41** valve inlet **42** aperture **43** manifold inlet **44** manifold outlet **50** motor **51** motor output **52** belt **55** motor mount **60** elbow **70** flow of liquid **75** denser material **100** electrical connector **105** main power switch **110** power supply **115** main logic controller **120** centrifuge motor variable frequency drive (VFD) **125** pump motor variable frequency drive (VFD) **130** pump motor **135** variable frequency drive (VFD) control signals **140** programming interface **145** auto/manual switch **150** centrifuge motor speed control **155** pump motor speed control **160** on switch **165** off switch **170** contamination sensors **175** pressure sensors **180** speed sensors **185** remotely-operated release valve

1. DESCRIPTION OF THE DEVICE

(11) The best mode for carrying out the device is presented in terms of its preferred embodiment, herein depicted within FIGS. **1** through **7**. However, the device is not limited to the described embodiment, and a person skilled in the art will appreciate that many other embodiments of the device are possible without deviating from the basic concept of the device and that any such work around will also fall under scope of this device. It is envisioned that other styles and configurations

of the present device can be easily incorporated into the teachings of the present device, and only one (1) particular configuration shall be shown and described for purposes of clarity and disclosure and not by way of limitation of scope.

(12) The terms “a” and “an” herein do not denote a limitation of quantity, but rather denote the presence of at least one (1) of the referenced items.

(13) The present device describes a continuous clean centrifuge **10**, which provides a cleaning means to a flow of liquid **70** passing therethrough. The continuous clean centrifuge **10** includes a manifold assembly **30** in fluid communication with a continuous flow of liquid **70** between a source of the flow of liquid **70** and downstream usage of a cleaned flow of liquid **70**. The continuous clean centrifuge **10** performs the cleaning means. In an exemplary method of such a cleaning means, the continuous clean centrifuge **10** has a motor **50** that rotates the flow of liquid **70** traveling through the manifold assembly **30** at a variable rotational rate relative to a fixed frame assembly **20**. This is accomplished by utilizing a logic program configured to operate and control the motor **50** generally between a faster, first rotation rate **11** and a slower, second rotation rate **12**, although incremental rates are envisioned to fall under the scope of the device. At the first rotation rate **11**, the continuous clean centrifuge **10** condenses denser material **75** from the flow of liquid **70** and at the second rotation rate **12**, the continuous clean centrifuge **10** discharges the denser material **75**. The flow of liquid **70** exiting the continuous clean centrifuge **10** is thus considerably cleaner than the flow of liquid **70** entering the continuous clean centrifuge **10**. These scenarios are depicted in FIGS. **6***a*-**6***b*.

(14) The motor **50** is controlled by a main logic controller **115**, which provides variable speeds thereto due to pre-programmed controls based on inputted parameters for providing the cleaning means to the flow of liquid **70**. The parameters may be dependent on the type of service that the continuous clean centrifuge **10** is placed in, the relative size of the denser materials **75** desired to be filtered or otherwise removed, the rate of filtering the denser materials **75**, etc. The service that the continuous clean centrifuge **10** can be placed in could include the production and handling of swimming pool water, wine, beer, sap, oil or wherever such cleaning means includes collecting and removing denser materials **75** from a flow of liquid **70** is desired or required. The continuous clean centrifuge **10** can also handle condensing and removal of water (as the denser material **75**) from a flow of compressed air (the flow of liquid **70**). The continuous clean centrifuge **10** may be scalable to handle up to fifty million gallons per day (50 M GPD).

(15) FIG. **1** illustrates the continuous clean centrifuge **10**, which includes a shroud **15** removably attachable to a frame assembly **20** that supports and protects the manifold assembly **30**. The frame assembly **20** comprises interlocking or mutually couplable frame elements that when constructed, can provide an open-faced rectangular prism structure, capable of generally guarding the manifold assembly **30** from or to the environment during operation. The shroud **15** can include a shroud front panel **16** removably attached to a front of the frame assembly **20**, a shroud rear panel **17** removably attached to a rear of the frame assembly **20**, a shroud first side panel **18** removably attached to a first side of the frame assembly **20**, and a shroud second side panel **19** removably attached to a second side of the frame assembly **20**. Any of the shroud panels **16**, **17**, **18**, **19** may have cut-outs to provide passage of elements of the manifold assembly **30**. The shroud **15** is generally used to provide noise, vibrational, and environmental protection to the continuous clean centrifuge **10** and may be removable and shall not affect any operation thereof.

(16) The frame assembly **20**, with or without the shroud **15**, can be removably or fixedly secured to a base plate **13**. The motor **50** is fixedly or removably mounted to the base plate **13** with a motor mount **55** and is in mechanical communication with the manifold assembly **30** such that it can rotatably motion the manifold assembly **30** to a rotational state when activated. This can be at pre-set or pre-determined revolution per minute (rpm) settings corresponding to the first rotation state **11**, the second rotation state **12**, or any incremental value thereabout of the rotational rate. Such a mechanical communication can be with a belt **52** transferring power from a motor output **51** of the

motor **50** to the manifold assembly **30**.

(17) Referring now to FIGS. **2-3**, it can be seen in various illustrations of the continuous clean centrifuge **10** that the manifold assembly **30** is rotatably movable relative to the frame assembly **20**. The frame assembly **20** further comprises a first side upper horizontal segment **21***a* and a second side upper horizontal segment **21***b* oriented parallel therewith. The first side upper horizontal segment **21***a* and second side upper horizontal segment **21***b* have a coextensive length. The frame assembly **20** also includes a front upper horizontal segment **22***a* and a rear upper horizontal segment **22***b* oriented parallel therewith. The front upper horizontal segment **22***a* and rear upper horizontal segment **22***b* have a coextensive length shorter than the first side upper horizontal segment **21***a* and second side upper horizontal segment **21***b*.

(18) The frame assembly **20** further comprises a first side lower horizontal segment **23***a* and a second side lower horizontal segment **23***b* oriented parallel therewith. The first side lower horizontal segment **23***a* and second side lower horizontal segment **23***b* have a coextensive length. The frame assembly **20** also comprises a front lower horizontal segment **24***a* and a rear lower horizontal segment **24***b* oriented parallel therewith. The front lower horizontal segment **24***a* and rear lower horizontal segment **24***b* have a coextensive length shorter than the first side lower horizontal segment **23***a* and second side lower horizontal segment **23***b*.

(19) A first side front vertical segment **25***a* has an upper terminal end couplable to the front terminal end of the first side upper horizontal segment **21***a* and the first side terminal end of the front upper horizontal segment **22***a*. The first side front vertical segment **25***a* has a lower terminal end couplable to the front terminal end of the first side lower horizontal segment **23***a*, and the first side terminal end of the front lower horizontal segment **24***a*. A second side front vertical segment **26***a* has an upper terminal end couplable to the front terminal end of the second side upper horizontal segment **21***b* and the second side terminal end of the front upper horizontal segment **22***a*. The second side front vertical segment **26***a* has a lower terminal end couplable to the front terminal end of the first side lower horizontal segment **23***b* and the second side terminal end of the front lower horizontal segment **24***a*. A first side rear vertical segment **25***b* has an upper terminal end couplable to the rear terminal end of the first side upper horizontal segment **21***a* and the first side terminal end of the rear upper horizontal segment **22***b*. The first side rear vertical segment **25***b* has a lower terminal end couplable to the rear terminal end of the first side lower horizontal segment **23***a* and the first side terminal end of the rear lower horizontal segment **24***b*. A second side rear vertical segment **26***b* has an upper terminal end couplable to the rear terminal end of the second side upper horizontal segment **21***b* and the second side terminal end of the rear upper horizontal segment **22***b*. The second side rear vertical segment **26***b* has a lower terminal end couplable to the rear terminal end of the second side lower horizontal segment **23***b* and the second side terminal end of the rear lower horizontal segment **24***b*.

(20) Auxiliary frame segments are further couplable to intermediate locations about the frame assembly **20**, mainly to counterbalance weight and positioning of sub-assemblies associated plumbing, logic control stations, or other environmental features. A first side front intermediate vertical segment **27***a* is attached at terminal ends to facing sides of the first side upper horizontal segment **21***a* and the first side lower horizontal segment **23***a* such that it is parallel with the first side front vertical segment **25***a* and first side rear vertical segment **25***b*. The first side front intermediate vertical segment **27***a* is located closer to the first side front vertical segment **25***a*. A first side rear intermediate vertical segment **27***b* is attached at terminal ends to facing sides of the first side upper horizontal segment **21***a* and the first side lower horizontal segment **23***a* such that it is parallel with the first side front vertical segment **25***a* and first side rear vertical segment **25***b*. The first side rear intermediate vertical segment **27***b* is located closer to the first side rear vertical segment **25***b*.

(21) A second side front intermediate vertical segment **28***a* is attached at terminal ends to facing sides of the second side upper horizontal segment **21***b* and the second side lower horizontal

segment **23***b* such that it is parallel with the second side front vertical segment **26***a* and second side rear vertical segment **26***b*. The second side front intermediate vertical segment **28***a* is located closer to the second side front vertical segment **26***a*. A second side rear intermediate vertical segment **28***b* is attached at terminal ends to facing sides of the second side upper horizontal segment **21***b* and the second side lower horizontal segment **23***b* such that it is parallel with the second side front vertical segment **26***a* and second side rear vertical segment **26***b*. The second side rear intermediate vertical segment **28***b* is located closer to the second side rear vertical segment **26***b*. The first side front intermediate vertical segment **27***a* may be aligned with the second side front intermediate vertical segment **28***a*. Similarly, the first side rear intermediate vertical segment **27***b* may be aligned with the second side rear intermediate vertical segment **28***b*.

(22) Referring now more closely to FIGS. **4** and **5**, the manifold assembly **30** can be generally seen as a rectangular element capable of independently spinning relative to the frame assembly **20**. A first manifold housing **32***a* and a second manifold housing **32***b* are formed as mirror images of each other, having coextensive perimeter edges and thicknesses. FIGS. **4** and **5** illustrate the second manifold housing **32***b* but it is understood that the first manifold housing **32***a* shall be formed in the mirror image. When inner faces of the first manifold housing **32***a* and second manifold housing **32***b* are mated together, a manifold **40** and any valve inlet **41** are formed as cylindrical conduits, as well as aligned apertures **42**, as well as a manifold inlet **43** and a manifold outlet **44**. The manifold **40** may meander in a serpentine pathway between the manifold inlet **43** and manifold outlet **44**. The manifold inlet **43** is in fluid communication with a first rotary coupler **35** and the manifold outlet **44** is in fluid communication with a second rotary coupler **46**. The manifold **40** is also in fluid communication with any valve inlets **41**. The valve inlets **41** are each in fluid communication with a remotely-operated release valve **185**. The valve inlets **41** and remotely-operated release valves **185** are preferably positioned on opposing and aligned sides of the manifold assembly **31**. In certain embodiments, plumbing in the form of elbows **60** may be in fluid communication between an individual valve inlet **41** and a respective remotely-operated release valve **185**. The aligned apertures **42** provide a secondary fastening means to secure the first manifold housing **32***a* and second manifold housing **32***b* together. The first manifold housing **32***a* and second manifold housing **32***b* may be constructed out of stainless steel piping, although polycarbonate plastic or other durable and resilient material may be utilized, as well as plain stock being bored or formed in situ. In an exemplary embodiment, the overall dimensions of the manifold assembly **31** are eight inches (8 in.) in width by twelve inches (12 in.) in length and one and one-half inches (1½ in.) in thickness. When fully formed, the diameter of the manifold **40** is one inch (1 in.).

(23) A flow of liquid **70** is introduced from a source to the first rotary coupler **35**, where it is transferred to the manifold **40** through the manifold inlet **43**. The cleaner flow of liquid **70** exits the manifold **40** through the manifold outlet **44** into the second rotary coupler **36** and delivered for downstream usage. Any flow of liquid **70** with denser material **75** is collected for subsequent removal via remotely-operated release valves **185**. Referring now back to FIGS. **1-3**, a first bearing **37** circumscribes the first rotary coupler **35** and is mounted to the first side of the frame assembly **20** and a second bearing **38** of the second rotary coupler **36** is mounted to the second side of the frame assembly **20**. Specifically in the exemplary embodiment, the first bearing **37** is mounted to the first side front vertical segment **27***a* and first side rear intermediate vertical segment **27***b* and the second bearing **38** is mounted to the second side front vertical segment **28***a* and second side rear vertical segment **28***b* such that the first rotary coupler **35** is positioned external of the frame assembly **20** and the first side rear intermediate vertical segment **29***b* and the shaft outlet **36** is positioned external of the frame assembly **20**. The communication between the first rotary coupler **35** and first bearing **37** and the communication between the second rotary coupler **36** and second bearing **38** enable the relative rotating motion of the manifold assembly **30** relative to the frame assembly **20**. A sprocket **39** circumscribes the first rotary coupler **35** at a position outside of the frame assembly **20**, preferably immediately adjacent to the motor output **51** of the motor **50**. The

motor **50** is in operable communication with the sprocket **39** via the belt **52** such that a rotational force generated by the motor **50**, when activated, is transferred to the motor output **51**, belt **52**, sprocket **39** and then transferred to the first rotary coupler **35** to enable rotation of the manifold assembly **30** relative to the frame assembly **20**.

(24) When the continuous clean centrifuge **10** is placed in the first rotation state **11**, as depicted in FIG. **6***a*, any denser material **75** is directed to an inner surface of outer portions of the manifold **40** (i.e., towards the perimeter edge) due to centrifugal forces acting on the flow of liquid **70**. These locations are advantageously aligned with a valve inlet **41**. In the first rotation state **11**, the remotely-operated release valves **185** are closed. When the continuous clean centrifuge **10** is placed in the second rotation state **12**, the remotely-operated release valves **185** are opened, such that a flow of liquid **70** with a very high concentration of denser material **75** can be delivered downstream for collection or treatment. The first rotary coupler **35** and second rotary coupler **36** have effective seals to seal the manifold inlet **43** and manifold outlet **44**, respectively, from any flow of liquid **70** or denser material **75** from leaking into the environment.

(25) A pump may be also mounted to the base plate **13** to be in fluid communication with the continuous clean centrifuge **10** such that it is able to deliver an amount of fluid thereto. The pump is operated by a pump motor **130** in operational and electrical communication with the main logic controller **115**. The pump is designed to deliver an amount of fluid with enough pressure to overcome any increased gravitational forces produced by the motor **50** to the flow of liquid **70** within the manifold **40** of the manifold assembly **30**. Any existing actuated inlet valves, actuated outlet valves, or proportioning valves may all be configured to be controlled with the main logic controller **115**.

(26) As is depicted in FIG. **7**, an electrical schematic of logic control for operation of the continuous clean centrifuge **10**, along with an exemplary method of use, is disclosed herein. Electrical power for the continuous clean centrifuge **10** is provided through an electrical connector **100**, such as a power cord or through direct connected power. Overall power requirements would be dictated by the total power draw of the individual electrical loads added together. Power would then be routed through a main power switch **105** and onto a power supply **110**. The power supply **110** would then provide power to a main logic controller **115** such as AutomationDirect Click PLC®. Other types of main logic controllers **115** including but not limited to Allen-Bradley Micro800 Series®, Siemens S7-1200®, Unitronics Unistream®, or even a Raspberry Pi® with Remote I/O, may also be utilized. The specific use of any specific type of main logic controller **115** is not intended to be a limiting factor of the present device. Power from the power supply **110** is also routed to a centrifuge motor variable frequency drive (VFD) **120** and a pump motor variable frequency drive (VFD) **125** for controlling the speed and torque characteristic of the motor **50** and a pump motor **130** via variable frequency drive (VFD) control signals **135** connected as outputs from the main logic controller **115**. The centrifuge motor VFD **120** would allow for speed control of the motor **50** up to and including twenty thousand revolutions per minute (20,000 RPM), although pre-set values may be used and generally categorized as speeds corresponding to the first rotation state **11** and the second rotation state **12**. Incremental speed control is also envisioned. The pump motor VFD **125** would allow for operation of the pump motor **130** within a range of forty to two hundred pounds per square inch (40-200 psi). The pump motor VFD **125** may be determined by the rotation rate of the centrifuge motor VFD **120**.

(27) The main logic controller **115** is also provided with a programming interface **140** to allow for logic programming of the continuous clean centrifuge **10** to deal with specific applications. The programming interface **140** is envisioned to be an HMI (Human-Machine Interface), or equal. The main logic controller **115** is also provided with an auto/manual switch **145** to allow either for programming via the programming interface **140** in an auto mode, or manual operation via a centrifuge motor speed control **150** and a pump motor speed control **155**. On and off control is performed by an on switch **160** and an off switch **165**, respectively, which also function as inputs to

the main logic controller **115**. The continuous clean centrifuge **10** is equipped with a multitude of other sensorial input, including but not limited to: contamination sensors **170**, pressure sensors **175**, and speed sensors **180**. The use of any specific type or quantity of sensors is not intended to be a limiting factor of the present device. The location of the sensors **170**, **175**, **180** can be placed anywhere in-line with the continuous clean centrifuge **10**, although locations at the manifold outlet **44**, the second rotary coupler **36**, or any of the remotely-operated release valves **185** may provide the most beneficial "snapshot" of the overall operation. Any and all sensors **170**, **175**, **180** function as input to the main logic controller **115**.
(28) Upon satisfying the requirements of the internal logic, the main logic controller **115** provides an output signal to any remotely-operated release valves **185**. The remotely-operated release valves **185**, in an exemplary embodiment, may be a spring-operated poppet valve. The remotely-operated release valves **185** will work with the internal logic of the main logic controller **115** and control of the centrifuge motor VFD **120** and pump motor VFD **125** to assure proper operation of the continuous clean centrifuge **10**. It is noted that the remotely-operated release valves **185**, envisioned to approximate four (4) units per plumbing section, may be controlled by a wide variety of methods including but not limited to: a pneumatic cylinder-operated type and an electric motor drive with a screw mechanism. The remotely-operated release valves **185** are then placed in fluid communication with downstream functions, if necessary, and convey the denser material **75**. Such downstream functions may include purification, sanitation, destruction, reclamation, etc.
(29) In another embodiment, the continuous clean centrifuge **10** consists of the base plate **13**, the shroud **15** including the shroud front panel **16**, shroud rear panel **17**, shroud first side panel **18**, and shroud second side panel **19**, and the frame assembly **20** formed from the first side upper horizontal segment **21***a*, second side upper horizontal segment **21***b*, front upper horizontal segment **22***a*, rear upper horizontal segment **22***b*, first side lower horizontal segment **23***a*, second side lower horizontal segment **23***b*, front lower horizontal segment **24***a*, rear lower horizontal segment **24***b*, first side front vertical segment **25***a*, first side rear vertical segment **25***b*, second side front vertical segment **26***a*, second side rear vertical segment **26***b*, first side front intermediate vertical segment **27***a*, first side rear intermediate vertical segment **27***b*, second side front intermediate vertical segment **28***a*, and second side rear intermediate vertical segment **28***b*. The frame assembly **20** supports a manifold assembly **30** consisting of an upper manifold portion **32***a* and a lower manifold portion **32***b* joined together to form a manifold **40** having one or more valve inlets **41** positioned along its periphery, a manifold inlet **43**, a manifold outlet **44**, and a plurality of apertures **42** for alignment and fastening. The manifold inlet **43** is in fluid communication with a first rotary coupler **35**, and the manifold outlet **44** is in fluid communication with a second rotary coupler **36**, with both couplers supported rotatably by a first bearing **37** and a second bearing **38**, respectively, mounted to the frame assembly **20**. A sprocket **39** is affixed to the first rotary coupler **35** outside the frame assembly **20** and receives rotational input from a motor output **51** of a motor **50** via a belt **52**, with the motor **50** mounted to the base plate **13** by a motor mount **55**, thereby imparting rotation to the manifold assembly **30** relative to the frame assembly **20**. The manifold **40** internally conveys a flow of liquid **70** introduced through the first rotary coupler **35** and manifold inlet **43** and directs the flow through the manifold **40** towards the manifold outlet **44** and second rotary coupler **36** for downstream use. Denser material **75** separated from the flow of liquid **70** during rotation is directed toward valve inlets **41** aligned with remotely-operated release valves **185**, with optional elbows **60** connecting the valve inlets **41** to the release valves **185** to facilitate the removal of collected denser material **75** during operation in the second rotation state **12**. The continuous clean centrifuge **10** is electrically powered through an electrical connector **100** connected to a main power switch **105** and a power supply **110**, which distributes power to a main logic controller **115**, a centrifuge motor variable frequency drive (VFD) **120** controlling the motor **50**, and a pump motor variable frequency drive (VFD) **125** controlling a pump motor **130** for pressurizing the incoming flow of liquid **70**. Variable frequency drive control signals **135** from the main logic controller **115** regulate

the operation of the centrifuge motor VFD **120** and the pump motor VFD **125**, permitting fine speed and pressure adjustments according to programmed parameters. A programming interface **140** permits the main logic controller **115** to be configured for either automatic or manual operation, selectable via an auto/manual switch **145**. Manual adjustments are performed through a centrifuge motor speed control **150** and a pump motor speed control **155**, while an on switch **160** and off switch **165** allow for basic power control. Sensor arrays, including contamination sensors **170**, pressure sensors **175**, and speed sensors **180**, are installed in-line at key locations such as the manifold outlet **44**, second rotary coupler **36**, and release valves **185** to provide real-time operational feedback to the main logic controller **115**. In response to programmed conditions and sensor inputs, the main logic controller **115** governs the operation of the remotely-operated release valves **185** to selectively evacuate accumulated denser material **75**, thus maintaining a continuous cleaning operation. All of the foregoing mechanical, fluid, and electrical communications among the respective numbered components work cooperatively within the continuous clean centrifuge **10** to provide an integrated system capable of operating through the first rotation state **11** for separation and the second rotation state **12** for purging.
(30) The foregoing descriptions of specific embodiments of the present device have been presented for purposes of illustration and description. They are not intended to be exhaustive or to limit the device to the precise forms disclosed, and obviously many modifications and variations are possible in light of the above teaching. The embodiments were chosen and described in order to best explain the principles of the device and its practical application, to thereby enable others skilled in the art to best utilize the device and various embodiments with various modifications as are suited to the particular use contemplated.

Claims

1. A continuous clean centrifuge, comprising: a frame assembly; a base plate coupled to the frame assembly; a manifold assembly rotatably mounted within the frame assembly; a motor mounted to the base plate and operatively connected to the manifold assembly to rotate the manifold assembly; a first rotary coupler positioned at a first end of the manifold assembly and coupled to a manifold inlet; a second rotary coupler positioned at a second end of the manifold assembly and coupled to a manifold outlet; a belt operatively connecting a motor output of the motor to a sprocket mounted on the first rotary coupler; a main logic controller configured to control operation of the motor; a plurality of valve inlets in fluid communication with a manifold formed in the manifold assembly; and, a plurality of remotely-operated release valves each fluidly connected to a respective valve inlet; and, wherein the motor rotates the manifold assembly at a first rotation state and a second rotation state; wherein at the first rotation state, denser material separates from a flow of liquid traveling through the manifold and accumulates adjacent the valve inlets; and, wherein at the second rotation state, the remotely-operated release valves are opened to discharge the denser material from the manifold assembly.
2. The continuous clean centrifuge of claim 1, wherein the frame assembly comprises a plurality of upper horizontal segments, lower horizontal segments, vertical segments, and intermediate vertical segments forming a generally rectangular prism.
3. The continuous clean centrifuge of claim 2, wherein the frame assembly includes a first side upper horizontal segment, a second side upper horizontal segment, a front upper horizontal segment, and a rear upper horizontal segment.
4. The continuous clean centrifuge of claim 2, wherein the frame assembly further includes a first side lower horizontal segment, a second side lower horizontal segment, a front lower horizontal segment, and a rear lower horizontal segment.
5. The continuous clean centrifuge of claim 2, wherein the frame assembly includes a first side front vertical segment, a first side rear vertical segment, a second side front vertical segment, and a

second side rear vertical segment.

6. The continuous clean centrifuge of claim 2, wherein the frame assembly further includes a first side front intermediate vertical segment, a first side rear intermediate vertical segment, a second side front intermediate vertical segment, and a second side rear intermediate vertical segment.

7. The continuous clean centrifuge of claim 1, wherein the manifold assembly comprises an upper manifold portion and a lower manifold portion, the upper and lower manifold portions mating together to form the manifold, the valve inlets, the manifold inlet, and the manifold outlet.

8. The continuous clean centrifuge of claim 1, wherein a first bearing circumscribes the first rotary coupler and a second bearing circumscribes the second rotary coupler, each bearing being mounted to the frame assembly.

9. The continuous clean centrifuge of claim 1, wherein the motor is controlled by a centrifuge motor variable frequency drive (VFD) configured to adjust the rotational speed of the motor between the first rotation state and the second rotation state.

10. The continuous clean centrifuge of claim 1, wherein the main logic controller is configured to receive input from a programming interface allowing operation in either an automatic mode or a manual mode.

11. The continuous clean centrifuge of claim 10, wherein the automatic mode adjusts motor speed based on parameters selected at the programming interface.

12. The continuous clean centrifuge of claim 10, wherein the manual mode allows user control of the motor speed using a centrifuge motor speed control.

13. The continuous clean centrifuge of claim 1, further comprising a shroud removably attached to the frame assembly, the shroud including a shroud front panel, a shroud rear panel, a shroud first side panel, and a shroud second side panel.

14. The continuous clean centrifuge of claim 1, wherein the manifold is formed with a serpentine pathway between the manifold inlet and the manifold outlet.

15. The continuous clean centrifuge of claim 1, further comprising elbows connecting the valve inlets to the remotely-operated release valves.

16. The continuous clean centrifuge of claim 1, wherein each remotely-operated release valve is a spring-operated poppet valve.

17. The continuous clean centrifuge of claim 1, further comprising a pump mounted to the base plate and configured to deliver the flow of liquid into the first rotary coupler.

18. The continuous clean centrifuge of claim 17, wherein the pump is operated by a pump motor controlled by a pump motor variable frequency drive (VFD) regulated by the main logic controller.

19. The continuous clean centrifuge of claim 1, further comprising a plurality of sensors selected from the group consisting of contamination sensors, pressure sensors, and speed sensors, wherein the sensors are in communication with the main logic controller.

20. The continuous clean centrifuge of claim 19, wherein the main logic controller controls the actuation of the remotely-operated release valves based on data received from the sensors.